MINERvA OmniFold

Research catch-up with Ben

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2D validation, higher-dimensional inference, and full-event learning

10 October 2026

The project now has three distinct outcomes

2D reproduction	Central result established; uncertainty transfer and coverage unresolved
3D to 5D scalar studies	Specific recovery tests pass; later tests expose limits
Full-event PET studies	Better recovery on some simulated shapes; intervals fail calibration

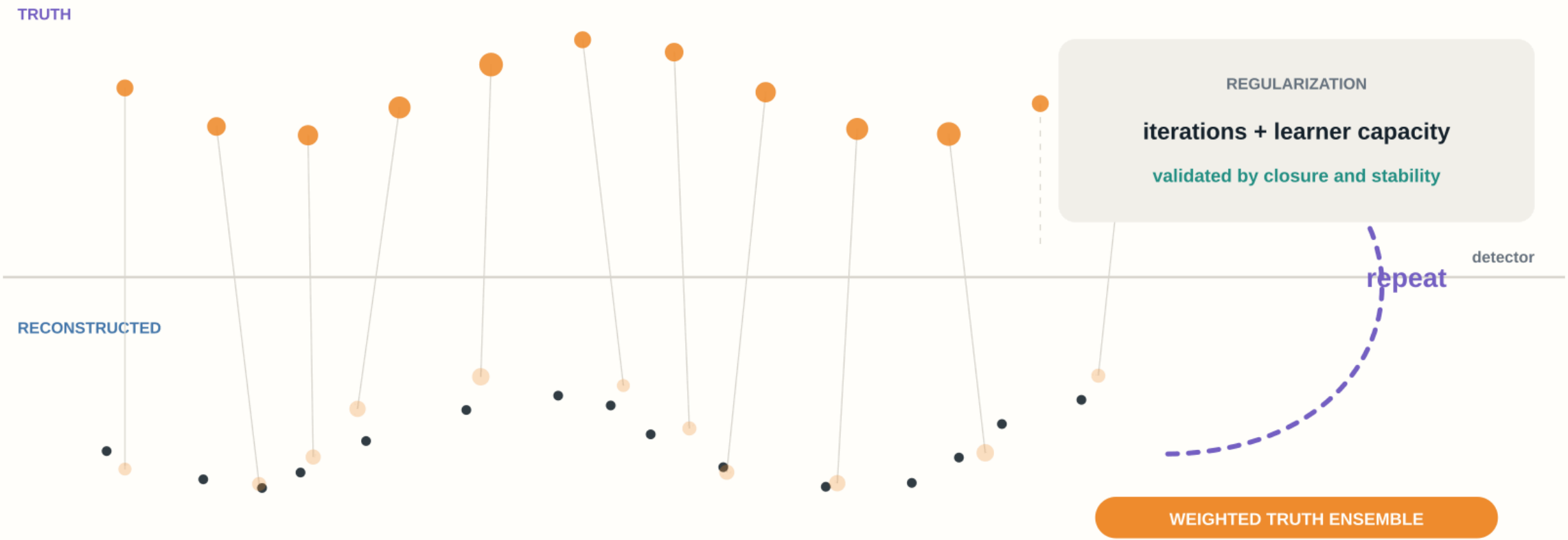
The data and the measurement

Sample	Medium-energy neutrino mode, about 6 GeV. 4.12 million selected events
Signal	Muon-neutrino charged-current interactions in the tracker
Reference domain	Muon angle $< 20°$, $1.5 < p_{\parallel} < 60$ GeV/c, $pT < 4.5$ GeV/c
Detector	MINERvA measures activity. Downstream MINOS measures the muon

THE TWO-STEP ALGORITHM

CONCEPTUAL TOY

One complete OmniFold iteration



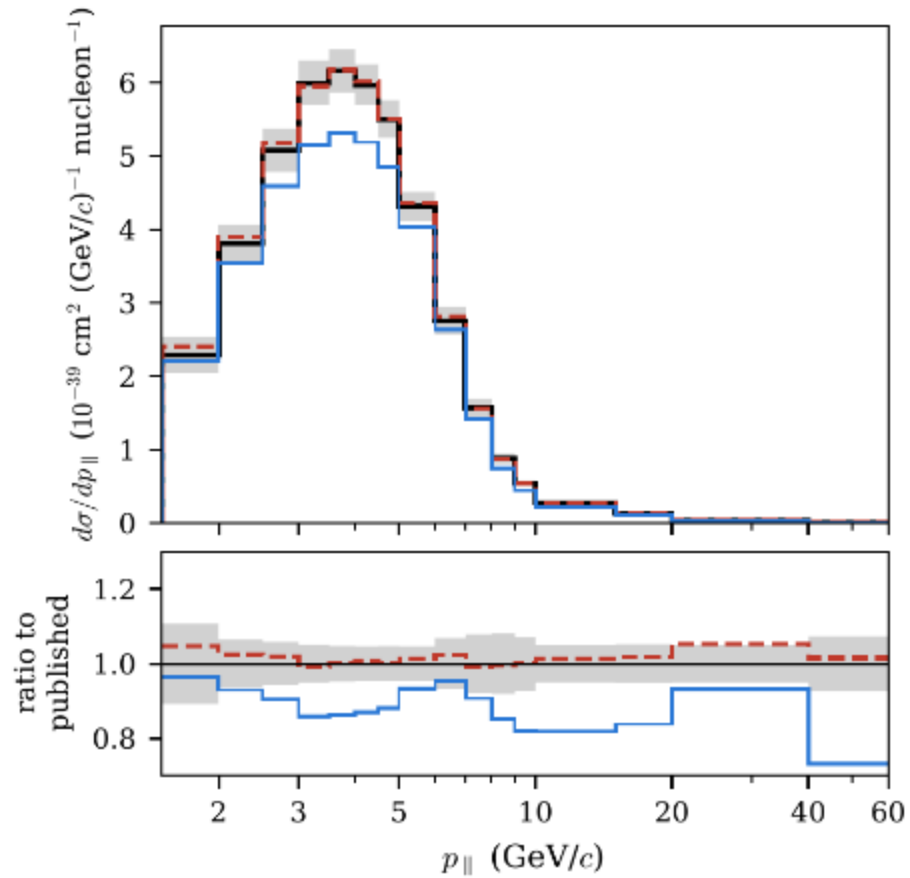
- After a controlled number of iterations, the weighted truth ensemble is the unfolded result.

Three different tests of an unfolding result

Closure	Create simulated data with known truth and test recovery
Dimensional anchor	Integrate an added observable and compare with the earlier result
Coverage	Repeat simulated experiments and count how often intervals contain truth

A closure pass applies to the tested deformation and detector response

The 2D cross section reproduces the published scale



Black: published Red dashed: OmniFold Blue: tuned prior

3.073 × 10^{-38} cm^2/nucleon

1.11% above the published total

205 reported bins. Five iterations

Same data and shared systematics

Projection of the validated 2D central result. This is a reproduction, not an independent measurement

The 2D controls test recovery and bookkeeping

Closure tests	Injected shapes: 0.046% / 0.013% median residual; limit 1.5%
Bookkeeping	Missed truth events preserve the full signal denominator
Iteration stability	5 → 10: total changes 0.026%; per-bin shape RMS 1.54%
Harder closure	Hidden-resolution amplitude 0.30: 3.43% residual; limit 3%

The 2D repair leaves two validation questions

Estimator pairing	Exact-split central; LightGBM uncertainties. Transfer unmeasured
Old band failed	91.21% containment; acceptable window 92.81–97.22%
Band repaired	Completeness fixed; adopted VL170 has no independent coverage re-test
Current status	Re-test deferred; Oct 7 diagnostic supports under-scattering of same-event toys

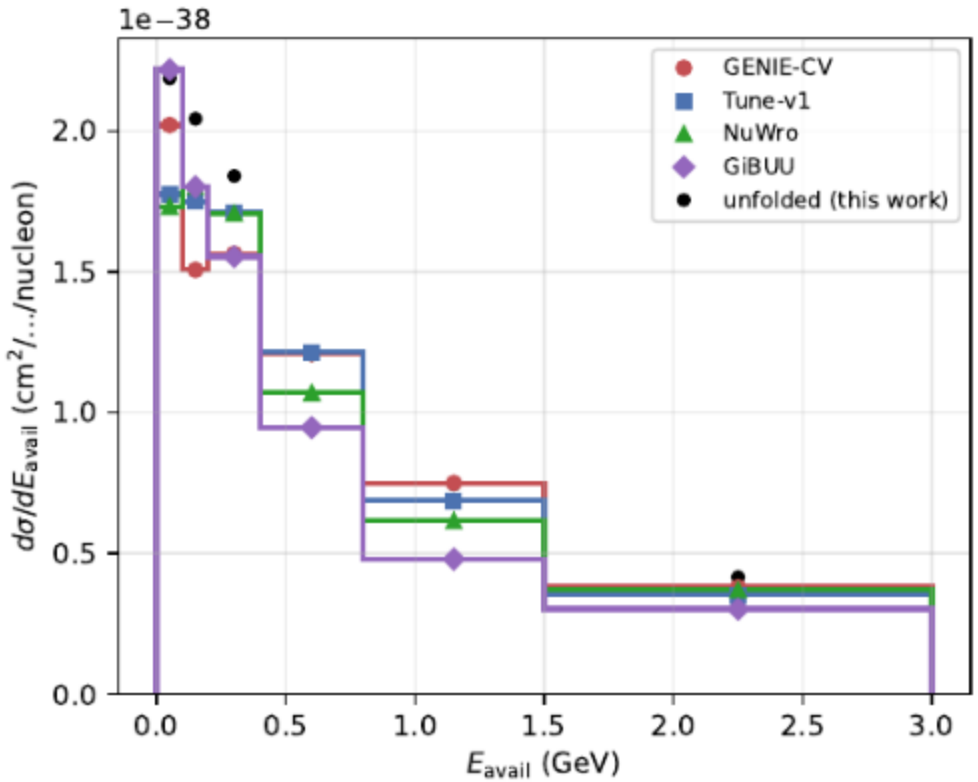
6.87% is the median bin budget against the LightGBM seed-42 central; not the error on the integral.

Each added observable asks a physics question

2D: pT, pI	Muon kinematics anchor the published measurement
3D: add Eavail	Visible hadronic energy resolves recoil structure
4D: add q3	Magnitude of momentum transfer adds interaction information
5D: add W	An inferred hadronic mass separates regions of the spectrum

Five recorded coordinates ≠ five independent degrees of freedom. Reco q3 and W share muon + recoil inputs.

Hadronic energy reveals model differences



3D normalization anchors to 2D

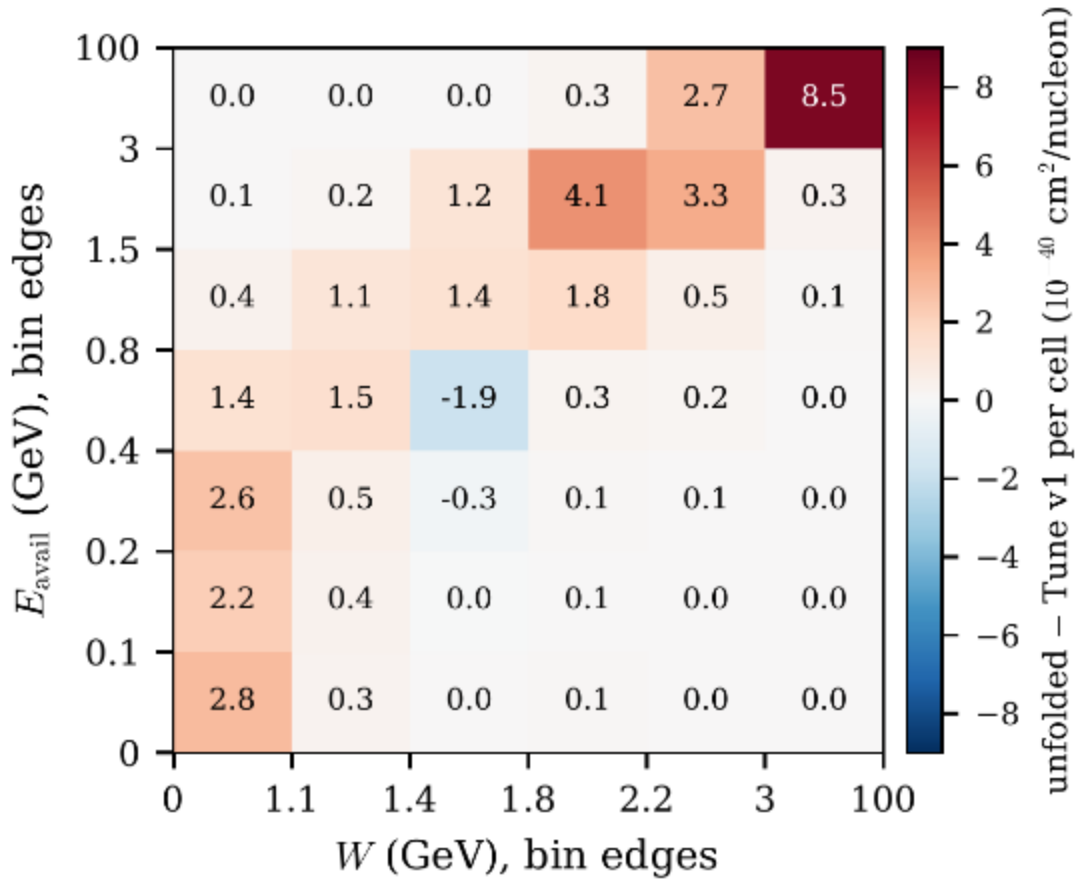
The tested Eavail injection closes

Valencia 2p2h fills 52% of the gap at Eavail ≤ 0.4 GeV

Central-value comparison only

Generator predictions use repaired samples. Historical covariance significances stay unquoted

The 5D projection locates a high-energy difference



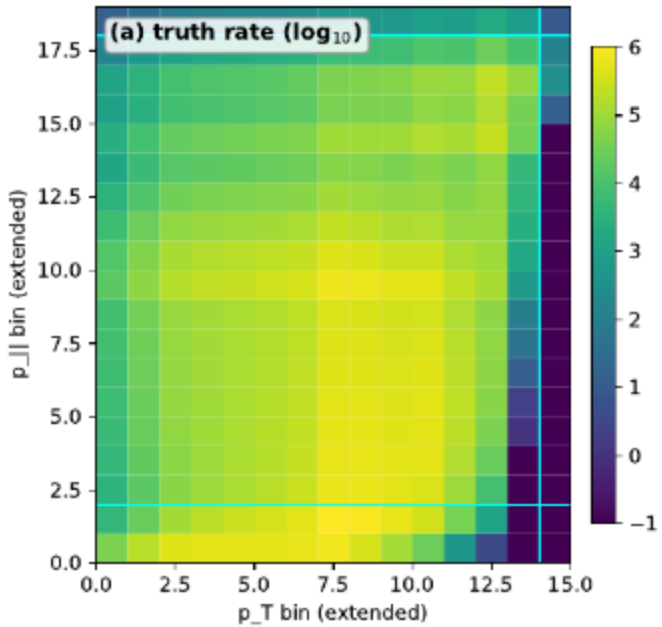
67% of positive excess at Eavail ≥ 0.8 GeV

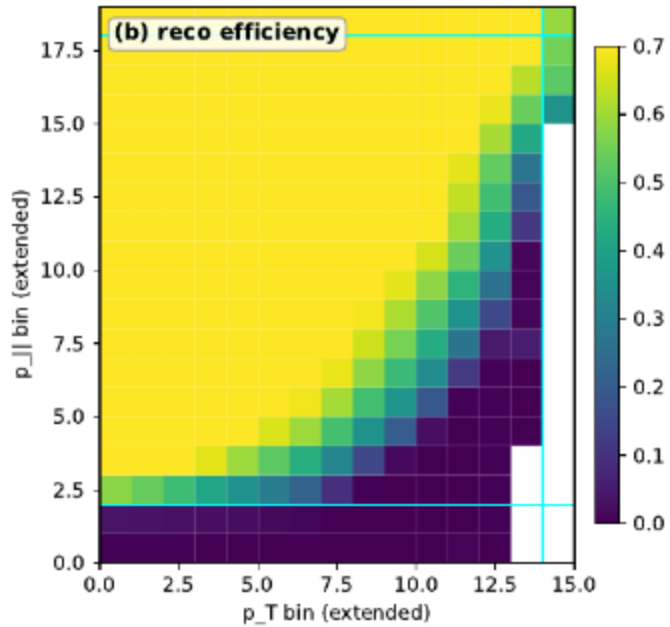
83% of that at W ≥ 1.8 GeV

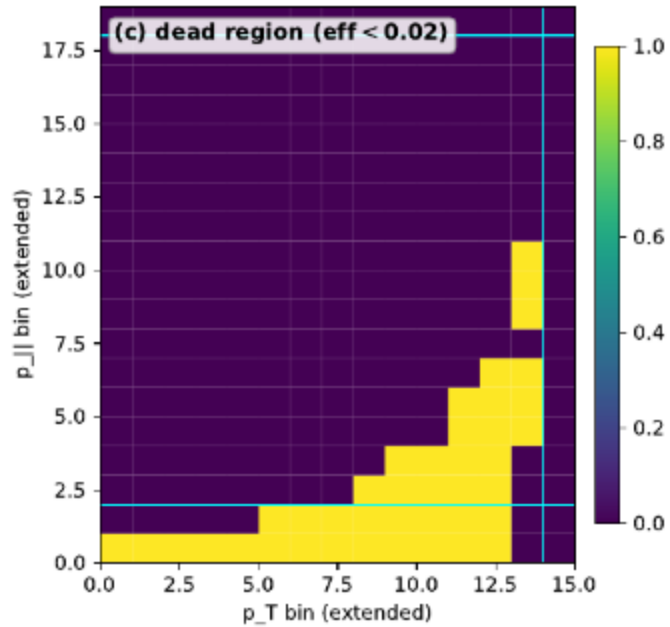
Largest cell: 22% of this map's positive excess

Cell-integrated central differences from MINERvA Tune v1. No ranking of significance across observables.

Extending the domain increases reliance on simulation







Remove truth-muon kinematic cuts

27.7% of truth rate lies where efficiency < 2%

Keep the reconstructed MINOS selection

Central-value and prior tests only

Extended fiducial domain. No corrected covariance for this extension

The adopted 5D covariance carries four measurements

1. Seed sensitivity	6.145% projected-width movement versus a 5% bound
2. Tested ensemble sizes	6.02%, 6.02%, 6.145% at N = 40, 80, 160. One seed pair
3. Unprobed bands	Five seed-pinned bands: 6.75% of trace. Direction of missing effect unknown
4. Central shifts	Up to 6.02% of projected uncertainty. Up to 49.8% in individual 5D bins

Digest 3d7465f6… adopted under a byte-specific exception. Product fields remain NON-PASSING / adoptable: false

The scalar campaigns exposed two distinct biases

Background treatment	Purity subtraction biases the highest-W column by about 4% in nominal toys
Truth departures	Generator-sized deformations leave residuals up to 74% in tested cells
Targeted repair	More refinement-classifier capacity repairs nominal background bias
Remaining failure	Candidate R fails real-data numerical stability. No estimator adoption

From an unqualified measurement to a model test

01

Assume a generator truth

Specify the fine-grid cross section and sub-grid model

This defines the null hypothesis

02

Simulate repeated experiments

Apply the declared nuisance variations and the estimator

Measure its behavior under that null

03

Compare the observed statistic

Count simulated tests at least as extreme

Correct for the ten comparisons

s5p measurement branch: NOT ADMITTED. The separate inference branch supplies no new reportable uncertainty

All ten tests reject their specified hybrid nulls

Five predictions × total and shape, familywise α = 0.05

1. Sub-grid shape is model-imposed. Refinement has not converged
2. Declared detector bands only. No recoil-energy-scale band
3. 34 interaction-model bands. Two negative-weight variations clipped
4. Half-MC toys versus full-MC data fit. Shift assessed and included
5. GiBUU has no Ev > 20 GeV. All predictions stop at 100 GeV
6. Generator finite-sample uncertainty enters the metric and draws

Conditional inference with candidate R, not a measured cross section. Independent recomputation agrees on the calculation

The inference has useful, measured limits

Grid refinement	Split cells more finely; κ scales the last observed change
Sensitivity scale	Primary κ = 2; robustness κ = 3; first loss ≈ 5.25 (frozen)
Muon-only comparison	Matched coarse muon projection does not reject GENIE CV
Limits	Recovered draws: ≈ 5.17. No convergence or complete response test

Muon comparison is post hoc. No qualified extra-power claim over all lower-dimensional tests.

PET gives OmniFold a set of objects per event

Detector objects

Energy deposits in scintillator views



Truth particles

Four-momenta and particle identity



Muon event features condition the network

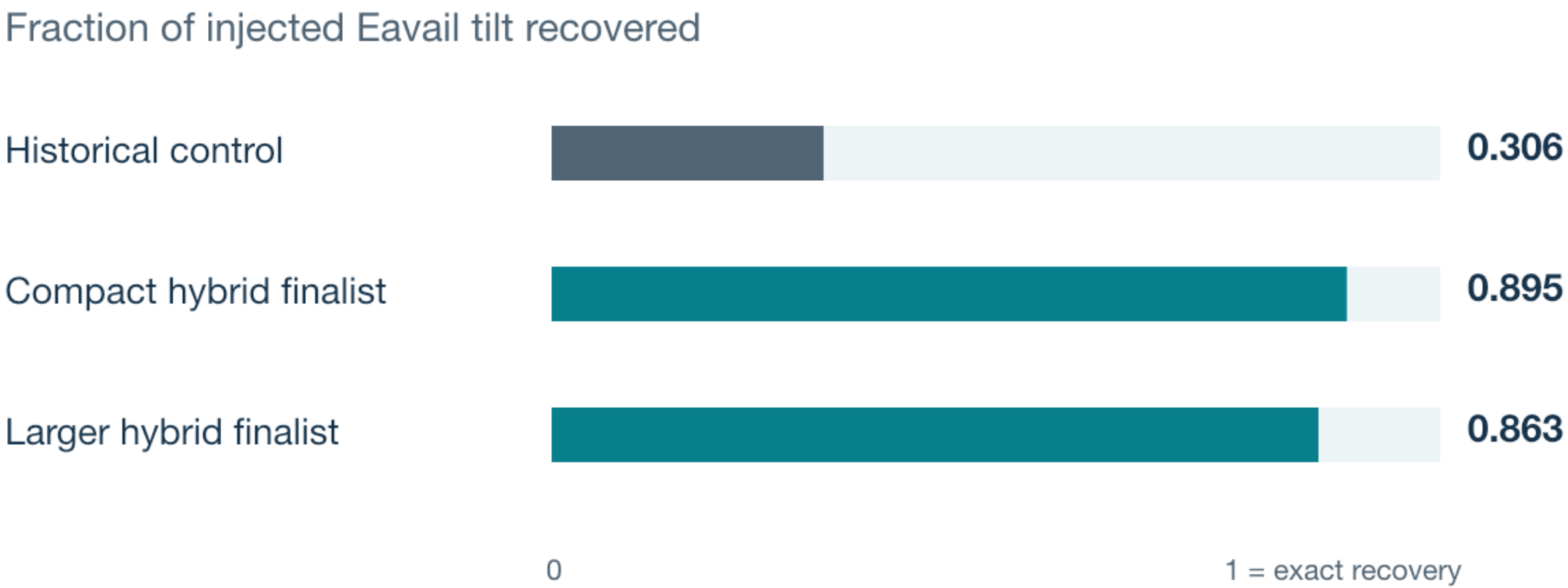
Attention learns object relationships; stored clouds retain at most 12 tokens

Historical recoil-only PET and the later full-event candidate have different input contracts

The earlier PET covariance did not qualify

Integral mismatch	Nominal / bootstrap mean = 1.1027 on 257 common cells
Localized mismatch	44 of 63 affected cells: nominal above all 50 replicas
Pairing declined	Median cell ratio 1.00032; mechanism unresolved
Scientific scope	Covariance unverified; no CML or adopted PET total covariance

PET recovery improved substantially in simulation



Final-bank Eavail tilt, 60 draws. R = 1 is exact recovery, R = 0 retains the prior discrepancy

Coverage counts intervals containing the true bin value

Region	Nominal 68%	Nominal 95%
Aggregate Eavail	89.6%	99.0%
Low acceptance	**25.7%**	**78.9%**
Moderate acceptance	70.1%	96.4%
Good acceptance	92.9%	99.8%

Too wide in aggregate, too narrow at low acceptance

120 pseudo-experiments. Entries pool bin–draw interval checks; these are not event-classification rates.

The paired PET–tree comparison depends on the distortion

Available-energy tilt	PET gains +0.102 / +0.070 over the paired tree comparator
Generator-shaped changes	Trees recover NuWro, NuWro′ and GiBUU reweightings better
Measured cost	PET: 1.768 / 1.517 A100-h per unfolding at declared packing
Tree campaign	352 tasks + control/retries: 5.306 charged CPU core-h total

Exploratory point-estimate comparison. Hardware hours are not a controlled speedup benchmark.

Reproducibility became part of the research

Execution integrity	An import-path defect could run another checkout's code despite matching file hashes
Scientific verification	Independent recomputation checks the calculation. It cannot validate every assumption
Stopping rules	Closed failed studies preserve useful limits and prevent retrospective selection
Deliverables	Note, primer, article and replay materials now separate claims by scope

Discussion with Ben

Article	Assembled, not submitted; publication-ready measurement still unmet
2D decision	Keep and disclose now; estimator/uncertainty transfer remains open
Proposed route	Matched LightGBM design with narrower claims; no adoption or run
Discussion	Which claim should lead, and what missing evidence matters most?

Backup: exact available-energy convention

Truth	Protons: E – 938.27 MeV. Charged pions: E – 135 MeV
Truth continued	Neutral pions and photons: total energy. All other species contribute zero
Reconstruction	1.17 × (tracker + ECAL blob energy after muon-fuzz subtraction)
Comparison limit	The inherited charged-pion constant and species exclusions differ from some published definitions

Backup: q3 and W are inferred kinematic quantities

q0	Energy transfer: neutrino energy minus muon energy at truth level
Q^2	Positive spacelike four-momentum-transfer variable
q3	Magnitude of momentum transfer: $q3^2 = Q^2 + q0^2$
W	Experimenter's mass: $W^2 = M^2 + 2M\ q0 - Q^2$

Backup: the adopted covariance's exact boundary

Identity	z-cv.npz, SHA-256 prefix 3d7465f6, 890,500,272 bytes
Four measurements	M1–M4 on slide 13 travel with every use of this product
Unresolved criterion	Cause 3 remains predeclared and not computed for this digest
Decision meaning	Exception covers these bytes only. No inherited adoption for other candidates

Backup: reading the conditional p-values

Finite simulation	$p = (k + 1) / (B + 1)$, with k exceedances in B toys
Resolution	B = 1,343–1,751. Never report p = 0 or turn it into a Gaussian significance
Test domain	109 supported joint cells. 72 for GiBUU with $p_{\parallel} < 6$ GeV/c
Missing draws	224 calibration + 53 power draws recovered. Frozen decisions unchanged

Backup: the PET representation studies

Pooling versus individual objects	Synthetic routing comparison found no resolved accuracy difference
Production cloud	The top-12 cap removes substantial reconstructed activity
Improvement campaign	No frozen ten-iteration candidate passed adequacy plus stress
Final design and pretraining	No tested PET2 recipe passed every development screen

Backup: the closed scalar sequence

s5c	Futility stop. Nominal purity-background bias confirmed
s5n	Refined subtraction improves nominal behavior. Departure failure persists
s5e	Diagnosed refinement underfitting. Candidate R fails fresh assessment
s5p	Measurement not admitted. Conditional inference completes with no new uncertainty

Backup: what the release can reproduce

Available	Article sources, analysis note, primer and saved inference inputs
Replay	Recompute conditional inference from sufficient products
Different task	Regenerate training and calibration pseudo-experiments end to end
Release limits	Several closure, uncertainty and sensitivity results remain outside RC4 replay

2D: what changed, and what is still open

Original reproduction	Truth/reco membership, missed events and normalization were repaired
October 6 repair	Bootstrap completeness made consistent with its central estimator
October 9 finding	Exact versus LightGBM central: median 0.97% per-bin difference
October 10 proposal	Matched estimator route; independent total coverage still not established

No revised central, uncertainty band or scientific gate is adopted by these design records.

Recorded coordinates and independent information

Muon	pT and p∥ determine momentum magnitude and polar angle
Transfer	$q3^2 = Q^2 + q0^2$; $W^2 = M^2 + 2Mq0 - Q^2$
Reconstruction	Both q3 and W use the same recoil q0 and muon inputs
PET / extended domain	More features or truth cells do not imply more identified directions

Equations describe the common kinematic picture; the code's truth/reco conventions and clamps differ.

More events: distinguish three different proposals

Same measured exposure	Current 2D uses all 12 ME-FHC playlists and about 4.12M selected events
Larger training sample	PET used subsamples; more MC can improve support and sampling precision
Different data / selection	RHC or new channels need their own flux, response and signal definition
Current uncertainty issue	More resampling does not create independent validation populations

Current MINERvA release lists ME FHC and RHC; low-energy availability should be checked before planning.